NexMart E-Commerce Sales Dashboard

Business Problem Statement & Project Deliverables

Dataset	Period	Records	Tools	Regions	Categories
NexMart Sales	2014–2017	9,994 Orders	MySQL + Power BI	4 US Regions	3 Product Lines

Business Problem Statement

NexMart, a multi-category e-commerce marketplace, has accumulated four years of transactional sales data across 9,994 orders spanning 4 major US regions and 3 product categories — Technology, Furniture, and Office Supplies. While the raw data exists, it remains unanalyzed, making it impossible for management to draw reliable conclusions or make data-driven decisions. The management team has observed fluctuating revenue patterns across different months, regions, and product sub-categories. They are unable to identify which customer segments generate the most value, which product lines are underperforming, or what geographic opportunities exist for targeted marketing and expansion.

> ***"How can NexMart leverage its 2014–2017 consumer sales data to identify high-value customer segments, optimize product category performance, and enable data-driven decisions across 4 US market regions?"***

Key Business Challenges

1

Raw Data Quality

9,994 rows with inconsistent DD-MM-YYYY date formats, column names with spaces and special characters, and unoptimized data types requiring thorough SQL-based cleaning.

2

Revenue Visibility

No consolidated view of total revenue ($2.30M), sub-category breakdown, or month-over-month trends — management cannot assess performance or identify seasonal demand patterns.

3

Customer Segmentation Gap

No system to classify customers into value tiers despite revenue variance ($14K–$25K). Top customer SM-20320 generates $25,043 across just 15 orders — a VIP opportunity missed.

4

Geographic Blind Spot

Sales across 4 US regions with no comparative analysis of region-level AOV or profit margin. West outperforms South by 85% in revenue ($725K vs $392K) — an unexplored opportunity.

5

Product Category Insight

Technology leads with Phones at $330K but Tables show -8.6% negative profit margin despite $207K in revenue. No cross-analysis exists between sub-categories and monthly trends.

Project KPI Snapshot

$2.30M	$286K	12.5%	793	9,994
TOTAL REVENUE	TOTAL PROFIT	PROFIT MARGIN	TOTAL CUSTOMERS	TOTAL ORDERS
Full Year 2014–2017	9,994 Total Records	Overall Blended	Unique Buyers	Validated Records

Project Deliverables

1 Data Cleaning & Preparation (MySQL)

- ✓ Loaded 9,994 rows into MySQL Workbench with correct data types
- ✓ Handled DD-MM-YYYY dates using STR_TO_DATE() for accurate parsing
- ✓ Used backtick quoting for columns with spaces and special characters
- ✓ Identified quality issues: spacing, Sub-Category dash formatting
- ✓ Output: superstore table — production-ready SQL dataset

2 SQL Query Analysis Engine (MySQL)

- ✓ Query 1 — Top Sub-Categories: Phones $330K, Chairs $328K (GROUP BY)
- ✓ Query 2 — Regional Performance: West $725K leads (HAVING clause)
- ✓ Query 3 — Customer LTV: SM-20320 with $25,043 lifetime value
- ✓ Query 4 — Monthly Trends: +71.6% MoM growth (LAG Window Function)
- ✓ Query 5 — Top per Category (ROW_NUMBER + PARTITION BY)
- ✓ Bonus — KPI Summary: Revenue, Profit, Margin, Customers, Orders

3 Advanced SQL Concepts Demonstrated

- ✓ GROUP BY + Aggregates: SUM(), COUNT(), AVG(), ROUND()
- ✓ HAVING clause: filtering aggregated results above $100,000
- ✓ Common Table Expressions (CTEs) using WITH clause
- ✓ LAG() Window Function for month-over-month growth calculation
- ✓ ROW_NUMBER() OVER (PARTITION BY) for top-per-group ranking
- ✓ STR_TO_DATE() + DATE_FORMAT() for custom date handling in MySQL

4 Interactive Power BI Dashboard

- ✓ Dark navy executive dashboard (#0F1117 / #181C2A background)
- ✓ 5 KPI cards: Revenue (blue), Profit (green), Margin (amber)
- ✓ Bar Chart: Top Sub-Categories by Revenue — from SQL Query 1
- ✓ Line Chart: Monthly Revenue Trend 2014–2017 — from Query 4
- ✓ Column Chart: Revenue by Region — from SQL Query 2
- ✓ Donut Chart: Top Sub-Category per Category — from Query 5
- ✓ Table: Top 10 Customers by Lifetime Value — from Query 3